

IndustrialBrain AI

AI-Powered Industrial Knowledge Platform

ET Hackathon 2026

Transforming industrial documents into actionable intelligence using AI, OCR, Knowledge Graphs and Enterprise Analytics.



Core Capabilities

- AI Chat
- OCR
- PDF Intelligence
- Knowledge Graph
- Analytics Dashboard

Presented By

Sai Kumar Kadiri

Aspiring Software Engineer

Java | Spring Boot | React



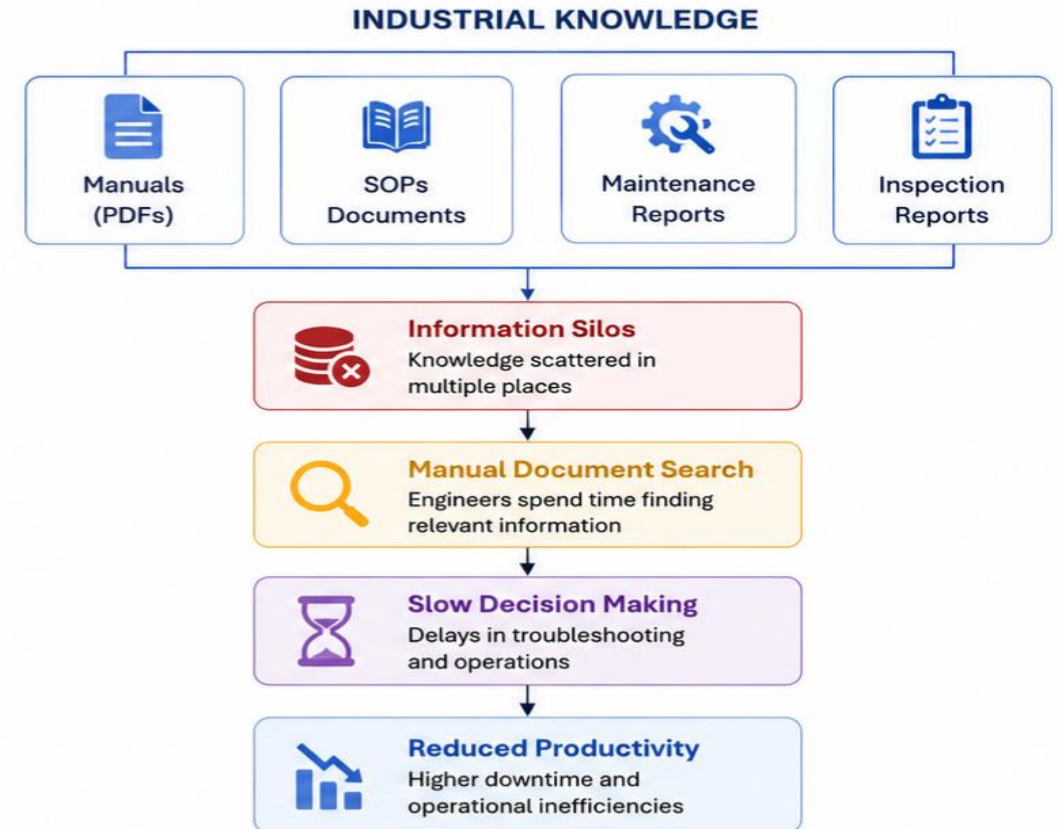
ET AI Hackathon 2026
AI Innovation Challenge

Problem Statement

Challenges in Industrial Document Management

Challenges in Industrial Document Management

- Large volumes of technical manuals and SOPs
- Information scattered across multiple PDF files
- Manual searching consumes engineering time
- Scanned documents require OCR before analysis
- No centralized knowledge repository
- Difficult to identify relationships between components

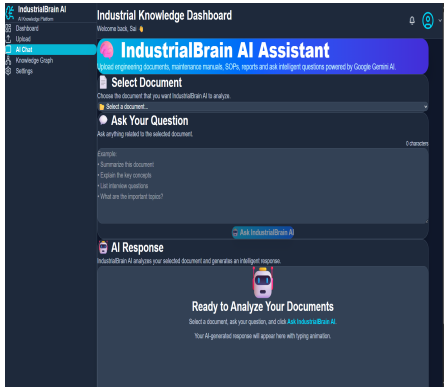
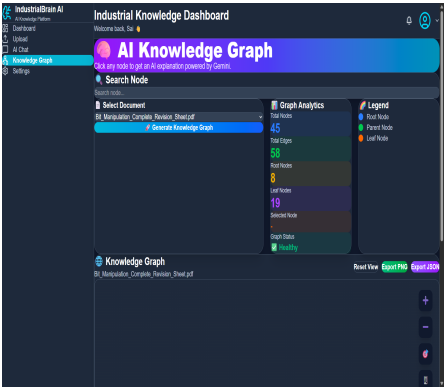
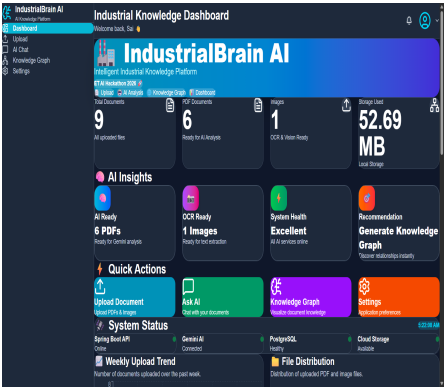
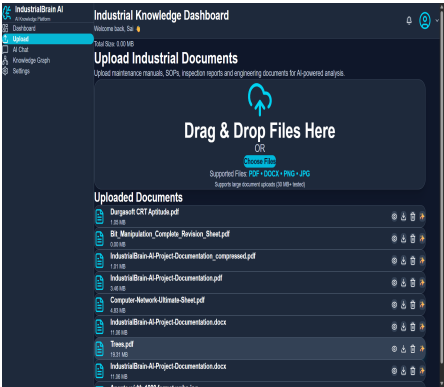
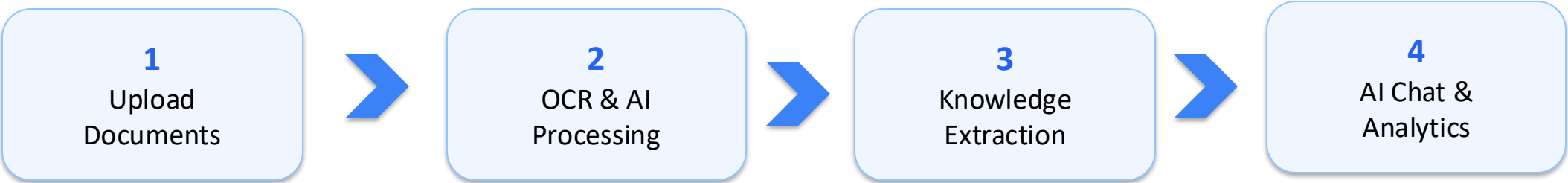


Why IndustrialBrain AI?

An enterprise AI platform that unifies OCR, AI chat, knowledge graphs, and analytics into a single intelligent workspace.

Solution Overview

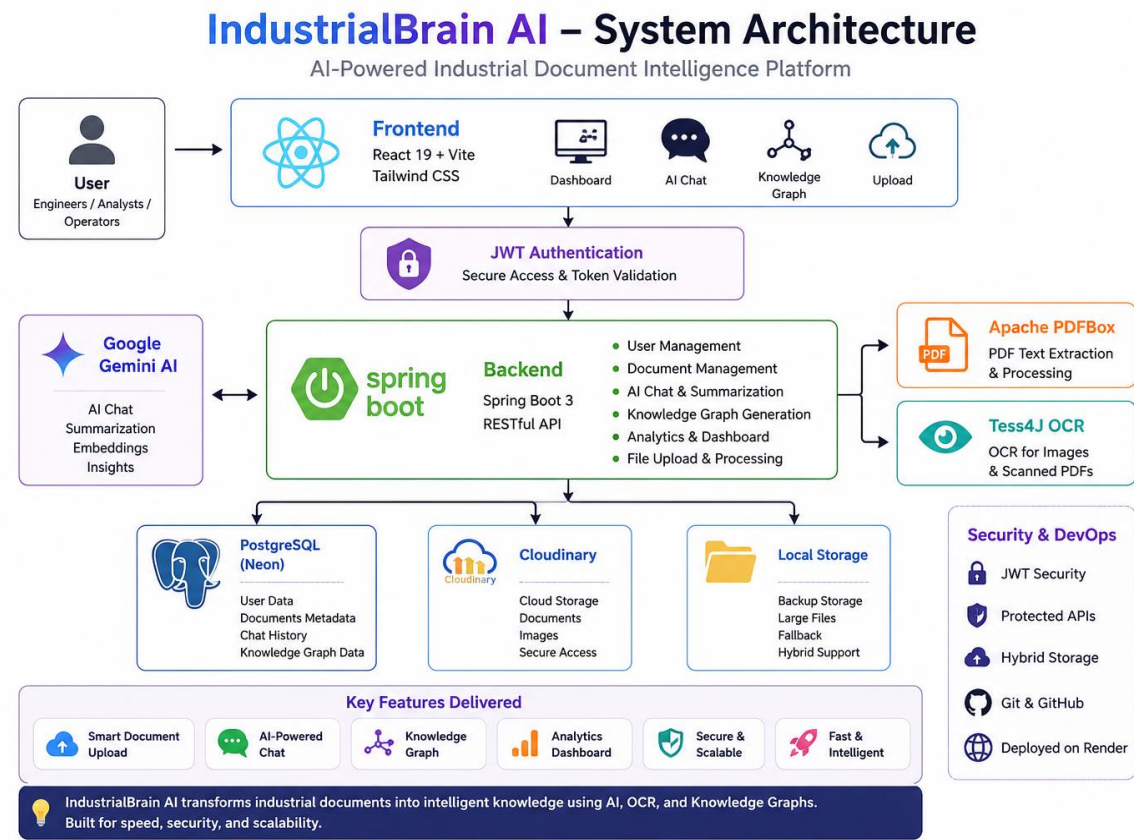
IndustrialBrain AI – End-to-End Intelligent Document Platform



IndustrialBrain AI transforms static industrial documents into an intelligent, searchable, AI-powered knowledge ecosystem.

System Architecture

Scalable Enterprise AI Architecture



Architecture Highlights: React Frontend • Spring Boot Backend • JWT Authentication • Google Gemini AI • OCR (Tess4J) • Neo4j Knowledge Graph • Cloudinary Storage • PostgreSQL (Neon) Database

Technology Stack

Enterprise technologies powering IndustrialBrain AI

Frontend

- React.js
- HTML5
- CSS3
- JavaScript

Backend

- Spring Boot
- Java
- REST APIs
- JWT

AI & NLP

- Google Gemini
- OCR (Tess4J)
- Prompt Engineering

Database

- PostgreSQL (Neon)
- Neo4j
- Cloudinary

Tools

- Git
- GitHub
- Postman
- Maven

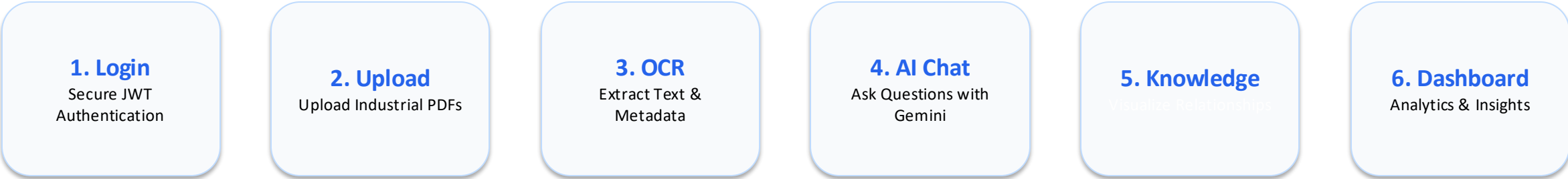
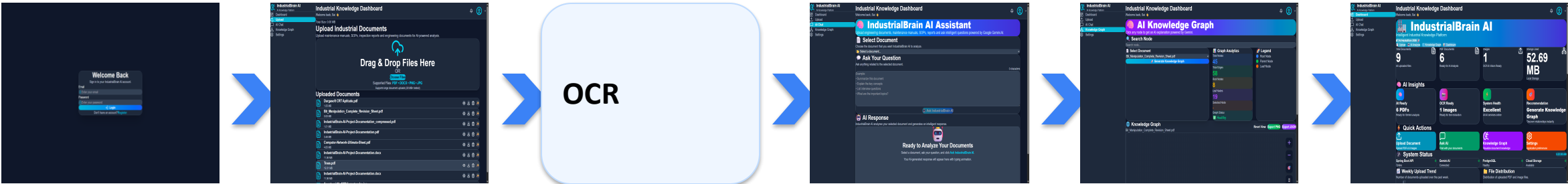
Deployment

- Vercel
- Render
- Docker Ready

Modern, scalable, cloud-ready architecture built with industry-standard technologies.

Application Workflow

End-to-End User Journey in IndustrialBrain AI



IndustrialBrain AI streamlines industrial knowledge management—from secure document upload to AI-powered insights and interactive knowledge exploration.

Dashboard & Analytics

Real-time insights and centralized document intelligence



 **Document Management**
View and organize uploaded industrial documents.

 **Analytics**
Track document statistics and processing status.

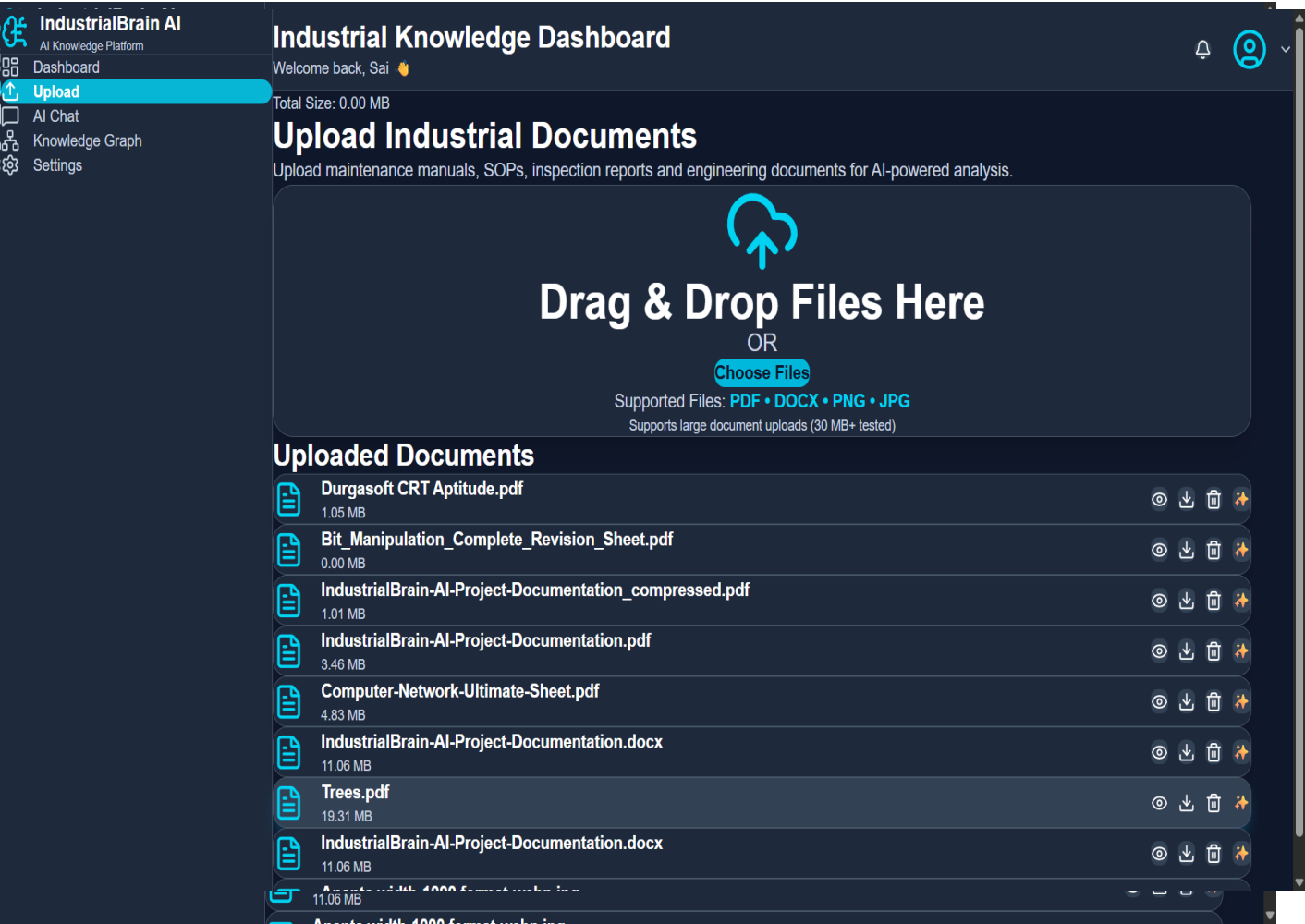
 **AI Insights**
Quick access to AI-powered summaries and recommendations.

 **Productivity**
Single dashboard for efficient industrial knowledge management.

Unified dashboard delivering analytics, AI intelligence, and operational visibility.

Document Upload & OCR

AI-powered document ingestion and intelligent text extraction



- ### 1. Secure Upload

Upload PDFs and scanned industrial documents.
- ### 2. OCR Processing

Extract text using Tess4J OCR.
- ### 3. AI Analysis

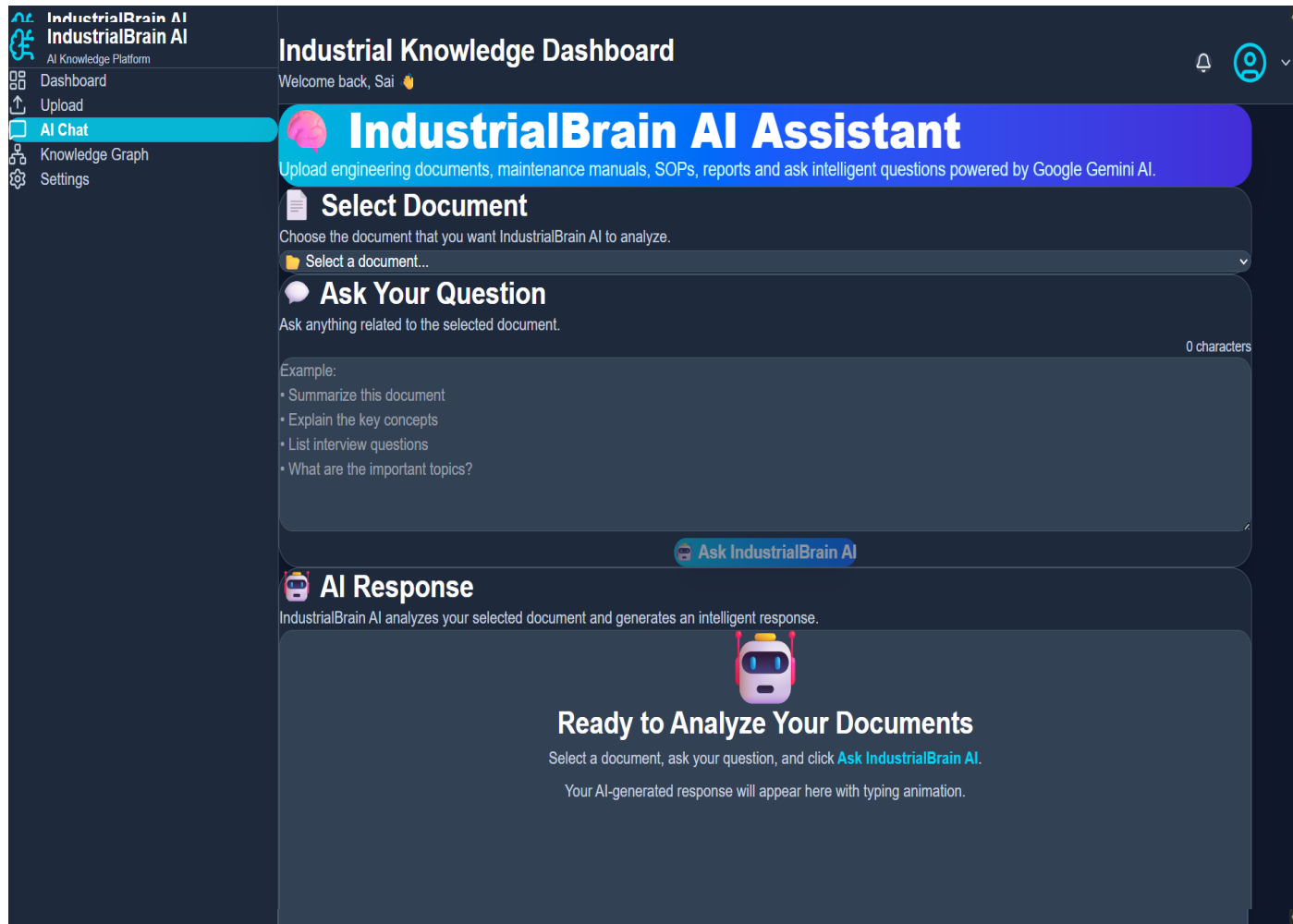
Google Gemini understands content and context.
- ### 4. Structured Output

Generate searchable knowledge for downstream modules.

From raw industrial documents to AI-ready knowledge in a streamlined workflow.

AI Chat Assistant

Context-aware document intelligence powered by Google Gemini AI



Natural Language Queries

Ask questions using everyday language without manual searching.

Context-Aware Responses

Gemini AI analyzes uploaded documents to deliver accurate answers.

Instant Knowledge Retrieval

Quickly locate procedures, specifications, and maintenance information.

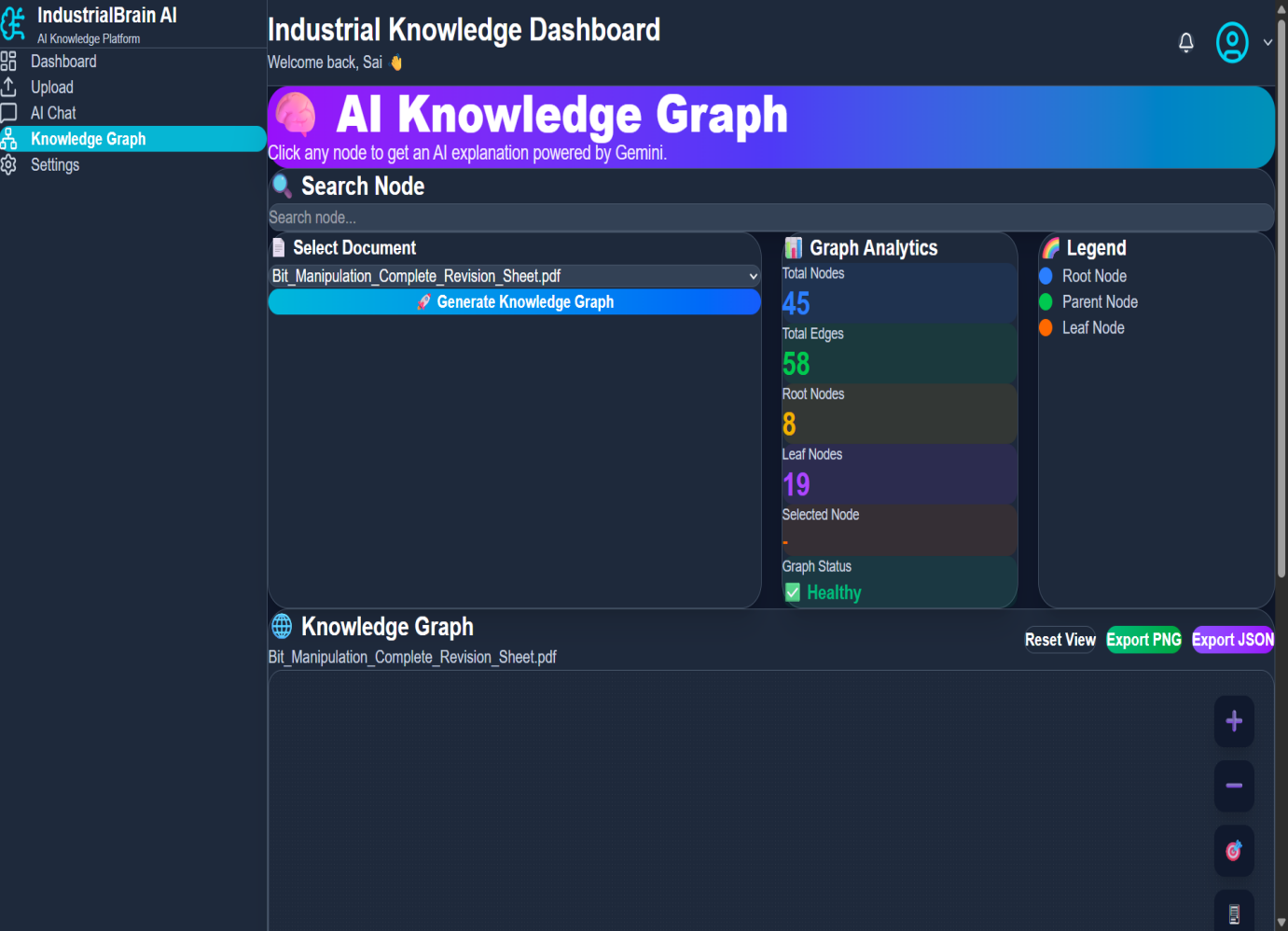
Decision Support

Accelerate troubleshooting and operational decision-making.

IndustrialBrain AI enables engineers to interact with complex industrial documentation through an intelligent conversational interface.

Knowledge Graph

Visualizing relationships between industrial entities using AI



Entity Extraction

Identify machines, components, procedures, and technical terms from documents.

Relationship Mapping

Connect entities to reveal meaningful operational relationships.

Interactive Exploration

Navigate interconnected knowledge for faster understanding.

Knowledge Discovery

Uncover hidden insights that support maintenance and decision-making.

Knowledge Graph transforms isolated documents into an interconnected network of industrial intelligence.

Security & Authentication

Enterprise-grade security for industrial knowledge management



JWT Authentication

Secure user login using JSON Web Tokens.



Protected REST APIs

Authenticated endpoints prevent unauthorized access.



Role-Based Access

Restrict access to sensitive industrial documents.



Secure Document Handling

Controlled upload and retrieval of PDF files.



Cloud Storage

Documents securely stored using Cloudinary integration.



Enterprise Ready

Scalable architecture following secure development practices.

Security-first design ensures confidentiality, integrity, and controlled access to industrial knowledge.

Business Impact & Key Benefits

How IndustrialBrain AI improves industrial knowledge management

Faster Information Retrieval

Engineers can locate relevant information in seconds instead of manually searching documents.

AI-Assisted Decision Making

Context-aware responses support maintenance, troubleshooting, and operations.

Centralized Knowledge

All industrial documents become a searchable, connected knowledge base.

Improved Productivity

Reduce repetitive work and increase operational efficiency.

Better Traceability

Knowledge Graph reveals relationships between assets, procedures, and components.

Scalable Enterprise Solution

Modern cloud-ready architecture supports future organizational growth.

IndustrialBrain AI transforms static documents into actionable intelligence, enabling faster, smarter, and more secure industrial operations.

Future Scope & Roadmap

Expanding IndustrialBrain AI into a next-generation enterprise intelligence platform

Phase 1	Multi-language document understanding
Phase 2	Voice-based AI assistant for engineers
Phase 3	IoT sensor integration & real-time monitoring
Phase 4	Predictive maintenance using AI/ML
Phase 5	Mobile application for field engineers
Phase 6	Cloud-native enterprise deployment & multi-tenant support

Our vision is to build a unified AI-powered industrial intelligence ecosystem for enterprises.

Thank You!

Questions?

Let's Build Smarter Industries with AI

Questions & Discussion

Thank you for your time and attention.

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Live Demo URL: <https://industrialbrain-frontend.onrender.com>



Scan to View GitHub
Repository